Climate Risk and Insurance Pricing in U.S. Long-Haul Trucking: A Simulation-Based Framework

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Abstract

Background. U.S. long-haul trucking is a corridor-dependent freight system whose insurance pricing has not been systematically linked to route-time climate exposure. Existing rating frameworks rely on static indicators including garaging territory, annual mileage, and historical loss experience that may fail to capture how climate variability shifts loss frequency, severity, and tail concentration.

Methods. We develop the Route-Time Climate Exposure Premium Indication Framework (RT-CEPIF), a two-part actuarial model that conditions claim frequency and claim severity on peril-specific weather states measured at the route-period level. A synthetic research-grade panel of 25,000 trips per seed is used to evaluate premium indication shifts across five climate scenarios: baseline, wet extreme, heat stress, flood tail, and adaptation.

Results. Flood-tail conditions raise mean indicated premium by 53.0% relative to baseline ($1,248 vs. $815). Wet-extreme and heat-stress scenarios generate increases of 13.7% and 10.6%, respectively. An adaptation scenario lowers premium by 5.1%. Weather-conditioned scoring more than doubles top-decile tail capture (0.44 vs. 0.15 under no-weather ablation) and materially reduces policy-month mean absolute error in every scenario.

Conclusions. Static rating structures appear systematically inadequate for climate-linked tail risk in commercial trucking. Dynamic, route-conditioned pricing that differentiates flood, precipitation, heat, and winter perils can improve premium adequacy, insurer capital management, and carrier adaptation incentives. Regulators should evaluate whether existing rate-review processes accommodate climate-sensitive segmentation.

Keywords: climate variability, long-haul trucking, insurance pricing, freight risk, actuarial modeling, route exposure, climate adaptation

Introduction

Long-haul trucking carries approximately 70% of the total value of freight shipped in the United States (FMCSA, 2007), and the primary routes it follows are concentrated across a limited set of interstate corridors. That geographic concentration creates operational efficiency but also systematic vulnerability: when weather disrupts a corridor, the consequences cascade across dispatch reliability, cargo integrity, driver safety, and insurer loss ratios simultaneously. Despite the scale of this exposure, commercial trucking insurance pricing has changed little in its fundamental structure. Most rating frameworks still depend on fleet size, garaging state, class code, and historical loss summaries, variables that aggregate exposure across space and time in ways that can obscure how route-time climate variability shifts underlying risk.

This paper argues that climate variability is not simply a background condition absorbed into historical averages. It is a structured, peril-specific driver of freight losses operating through identifiable channels: precipitation increases braking distance and hydroplaning risk; heat degrades tires, cooling systems, and temperature-sensitive cargo; flood events close corridors and concentrate physical losses in the tail; snow and cold affect visibility, stopping distance, and mechanical reliability under schedule pressure. Each channel produces a different operational signature and a different actuarial profile. Pricing that fails to differentiate these channels will systematically underprice tail risk, cross-subsidize high-exposure routes, and weaken the adaptation incentives that insurers are otherwise positioned to provide.

Several bodies of evidence motivate this concern. Research on single-vehicle truck crashes shows that wind speed, rain, humidity, and air temperature are each associated with injury severity in statistically distinct ways (Naik et al., 2016), and that weather conditions should be modeled separately for rain and snow rather than collapsed into a single inclement-weather indicator (Uddin and Huynh, 2020). The FMCSA crash causation literature documents how environmental conditions interact with driver behavior and vehicle characteristics in the generation of large-truck crashes (FMCSA, 2007, 2006). On the insurance side, recent work demonstrates that climate-related hazards measurably shift motor insurance claim frequency (Fung et al., 2023) and that freight insurance pricing can be substantially improved by incorporating route-level risk factors beyond traditional fleet descriptors (Yang et al., 2023). Marine cargo research finds a rising trend in weather-related claims over the past decade and identifies important gaps in insurer readiness for climate-linked supply chain disruption (du Plessis et al., 2024a,b).

We develop the RT-CEPIF framework that formalizes this logic as a two-part actuarial model, linking climate hazard states at the route-period level to claim frequency and claim severity, and then carrying those effects through a pricing equation that includes trend, development, capital load, and reinsurance load. The research is currently built around a synthetic but research-grade data architecture with a clear migration path to observed carrier and insurer data. The central evaluative claim is that weather-conditioned premium indication materially outperforms static alternatives on policy-month expected loss and stress-tail ranking, even where gains remain modest at the single-trip level.

Background and Literature

The effect of weather on transportation has been studied extensively in the transport literature (Koetse and Rietveld, 2009). Adverse weather increases travel time, raises accident rates, disrupts schedules, and amplifies infrastructure stress. Long-haul trucking is particularly sensitive because it combines commercial service obligations, federally regulated hours-of-service constraints, asset-intensive operations, and corridor-dependent routing. Route heterogeneity is therefore a structural feature of trucking risk: Gulf routes, Great Plains winter corridors, mountain passes, and desert heat belts present materially different hazard profiles, and a carrier domiciled in one state may generate most of its exposure on routes through another.

The trucking safety literature has made substantial progress in characterizing how specific meteorological conditions shape crash outcomes. Naik et al. (2016) combined 15-minute weather station data with crash and roadway records in Nebraska, finding that greater wind speed and rain are associated with more severe injuries in single-vehicle truck crashes while snowy and icy pavement conditions paradoxically reduce injury severity, an effect attributed to drivers slowing and heightening vigilance in winter conditions. Uddin and Huynh (2020) extended this approach using Ohio crash data from 2011 to 2015, demonstrating through parameter transferability tests that rain and snow generate statistically distinct

injury severity profiles; models pooled across weather conditions obscure the heterogeneous effects that disaggregated specifications reveal. These findings from the crash literature map directly onto actuarial implications: if rain and snow affect the severity of each loss event differently, they should enter premium indication through separate channels rather than a single weather indicator.

The FMCSA Large Truck Crash Causation Study provides the foundational empirical base for understanding how driver, vehicle, environment, and roadway factors combine in crash events (FMCSA, 2007). The Report to Congress on the LTCCS elaborates on the policy and safety implications (FMCSA, 2006), and the LTCCS data portal enables replication and route-level risk stratification for subsequent research (FMCSA, 2024). The crash causation evidence supports the view that driver fatigue, schedule pressure, and equipment condition interact with environmental hazards rather than acting independently, which motivates the operational mediation logic at the core of the RT-CEPIF framework.

The insurance literature has well-established methods for decomposing commercial auto losses into frequency and severity components and translating those components into rate indications through two-part actuarial models, generalized linear models, and credibility frameworks. Telematics-based underwriting has demonstrated that operational behavior captures information about loss risk beyond static policy descriptors (Baecke and Bocca, 2017). A closely related recent contribution by Fung et al. (2023) estimates that climate-related hazards produce detectable shifts in motor insurance claim frequency prediction when incorporated into count regression models, providing direct evidence that weather exposure belongs in actuarial pricing and not only in transportation planning. Yang et al. (2023) demonstrate on a Chinese freight platform dataset that incorporating purchase behavior and road density alongside traditional risk factors significantly improves claim probability fit in a generalized linear model framework, illustrating how digital platform data can support more granular freight insurance pricing.

On the cargo side, marine insurance research offers useful parallels. du Plessis et al. (2024a) examine insurer perceptions of climate change risk across an international cohort and find that higher-income countries tend to show greater awareness and more structured risk management practices, while gaps remain in the early identification and assessment steps of supply chain risk frameworks. A companion study (du Plessis et al., 2024b) documents a rising trend in weather-related marine cargo insurance claims from a South African perspective, emphasizing the financial performance implications for insurers that have not yet updated their pricing for climate-linked claim patterns. Taken together, this marine cargo evidence reinforces the concern that a passive approach to climate risk is insufficient in commercial lines of insurance.

On the insurance governance side, Kunreuther and Michel-Kerjan (2009) and Jaffee et al. (2010) have documented how rising climate-linked volatility creates pressure on pricing adequacy, capital requirements, and market availability. These dynamics are well understood in property catastrophe insurance but have received less attention in commercial trucking. The mechanisms are structurally similar: if exposures shift faster than rating systems, mispricing can persist until corrected by underwriting tightening, premium surges, or market exit. Understanding those mechanisms before they generate market disruption is the practical motivation for this paper.

Conceptual Framework

Causal Chain

Our framework represents the path from climate hazard to insurance pricing as a staged causal chain:

$$\text{climate hazard} \rightarrow \text{route-time exposure} \rightarrow \text{operational mediation}$$
$$\rightarrow \text{insured loss} \rightarrow \text{actuarial translation} \rightarrow \text{premium indication}$$

This chain matters because climate does not directly produce premium changes. Hazards must first be experienced by trucks on actual routes at actual times. Realized exposure is then mediated by dispatch pressure, driver behavior, equipment condition, maintenance quality, cargo sensitivity, deductibles, and coverage terms. Only after these steps do insurers observe loss outcomes that can update pricing or underwriting.

Static rating systems short-circuit this chain by substituting annual geographic summaries for route-time hazard matching. This may be adequate when climate variability is low and the correlation between policy domicile and realized route exposure is high. It becomes inadequate when extreme events are episodic, corridor-concentrated, or regime-shifting in ways that historical averages do not capture, precisely the conditions that the crash causation literature (FMCSA, 2007) and emerging climate-insurance research (Fung et al., 2023) suggest are becoming more common.

Peril-Specific Hypotheses

We advance six hypotheses that differentiate the causal pathways of distinct climate perils.

H1: Extreme precipitation increases claim frequency more than non-extreme wet conditions, through hydroplaning, visibility loss, and extended braking distance. This is consistent with the crash-severity evidence in Naik et al. (2016) and the claim-frequency evidence in Fung et al. (2023).

H2: Heat-wave exposure increases claim severity more than frequency, through cargo spoilage, tire failure, mechanical degradation, and pavement stress. The cargo vulnerability logic is supported by marine insurance evidence on temperature-sensitive goods (du Plessis et al., 2024b).

H3: Snow and cold effects are conditional on route type, fleet winterization, and dispatch pressure; adaptation reduces but does not eliminate winter weather effects. The paradox that snowy pavement can reduce injury severity (Naik et al., 2016; Uddin and Huynh, 2020) reflects driver behavioral adjustment, which the framework captures through operational mediation.

H4: Static rating variables systematically underprice climate-linked tail risk because garaging territory and lagged experience fail to capture route-time clustering of flood and storm exposure.

H5: Contract design and safety technology moderate the translation from weather exposure to insured loss; deductibles, telematics, and driver-assistance systems reshape observed outcomes (Baecke and Bocca, 2017).

H6: Apparent climate effects partially attenuate after controlling for route structure, underwriting composition, and operational selection. Carriers that reroute away from flood exposure may appear safer not because floods are harmless but because adaptation changed realized exposure.

Data and Methods

Research Design

This paper uses a structured synthetic empirical design. The design is not a substitute for real-world evidence; it is a disciplined architecture for testing model logic, variable construction, identification assumptions, and reproducibility before migration to confidential carrier and insurer data. The migration path is explicit: weather fields map to NOAA Climate Data Online gridded products (NOAA, 2025); route-time exposure maps to electronic logging device or GPS trip traces; insurance fields map to commercial auto policy, premium, and claims systems; and crash causation context maps to FMCSA public datasets (FMCSA, 2024).

The unit of analysis is the trip-policy observation, with claims linked one-to-many and aggregated to the policy-month level for pricing evaluation. Each simulation panel contains approximately 25,000 trips. Results are reported across three random seeds and five climate scenarios to characterize simulation uncertainty rather than a single deterministic outcome.

The RT-CEPIF Model

The RT-CEPIF framework is a two-part actuarial model. The frequency component is a negative binomial specification:

$$\log(\mu_{irt}) = \log(\text{Exposure}_{irt}) + \alpha_r + \tau_t + \beta^\top W_{irt} + \gamma^\top B_{irt} + \delta^\top K_{irt} + \psi^\top (W \times K)_{irt}$$

where r indexes route corridor, t indexes time period, W is a vector of peril-specific weather indicators, B captures operational behavior including hours-of-service pressure, maintenance risk, and advanced driver assistance, and K captures contract structure including deductible and coverage breadth. The severity component is a Gamma log-link model conditioned on positive claim occurrence, with an additional catastrophe branch for flood events:

$$\log(m_{irt}) = a_r + b_t + \theta^\top W_{irt} + \eta^\top B_{irt} + \zeta^\top K_{irt} + \rho^\top (W \times \text{CargoVulnerability})_{irt}$$

The expected pure premium is $\widehat{PP}_{irt} = \hat{\mu}_{irt} \times \hat{m}_{irt}$, and the premium indication is:

$$\widehat{PI}_{irt} = \widehat{PP}_{irt} \cdot T \cdot D + F + v\,\widehat{PP}_{irt} + CL_{irt} + RL_{irt}$$

where T is the trend factor, D is the loss development factor, F is fixed expense, v is the variable expense rate, CL is the capital load, and RL is the reinsurance load. Capital and reinsurance loads are explicitly functions of the tail-sensitive climate state, so that flood and extreme-weather regimes carry higher loads. This design echoes the two-part pricing logic in Yang et al. (2023) and the climate-conditioned frequency approach in Fung et al. (2023), generalized here to incorporate route-time peril exposure with operational and contract moderators.

The framework coefficients used in the scoring engine are reported in Table 1. These are calibrated scoring weights, not estimated standard-error confidence intervals from observed microdata. The peril-specific structure, with flood and snowstorm carrying the largest frequency-side loadings, is consistent with the injury-severity evidence from FMCSA (2007), Naik et al. (2016), and Uddin and Huynh (2020), which collectively document that these weather states generate the most consequential operational disruption in large-truck operations.

Climate Scenarios

Five scenarios are evaluated against a baseline configuration:

- **S0 Baseline**: ambient weather conditions calibrated to moderate U.S. corridor variability.
- **S1 Wet Extreme**: heavy precipitation frequency raised; flood probability elevated.
- **S2 Heat Stress**: sustained heat-wave exposure; cargo vulnerability amplified.
- **S4 Flood Tail**: flood-tail probability sharply elevated; catastrophe branch activated.
- **S5 Adaptation**: carrier-level winterization, routing flexibility, and telematics adoption improved.

Benchmark Models

Three benchmark models are evaluated: (B0) a constant-loss benchmark assigning mean loss to every trip; (B1) a static underwriting benchmark using fleet size, garaging state, and class code; and (B2) the RT-CEPIF climate-conditioned model. A no-weather ablation of RT-CEPIF, identical to the full model but with all weather channels zeroed, serves as the primary comparison for isolating the value of climate information. Evaluation targets are policy-month mean absolute error (MAE) and premium-ranked top-decile tail capture, which are actuarially meaningful rather than purely statistical.

Results

Scenario Premium Indication Shifts

Table 2 reports the main scenario results across three seeds on the small-panel configuration and one large-panel seed. The ordering is consistent across all specifications: flood-tail conditions produce the largest premium displacement, followed by wet extreme and heat stress, and adaptation lowers premium below baseline in all seeds.

On the distributed large-panel configuration (seeds 7, 11, 19), the mean indicated premium rises from \$828 at baseline to \$943 under wet extreme (+13.9%), \$919 under heat stress (+11.0%), and \$1,214 under flood tail (+46.6%). Adaptation reduces mean premium to \$784 (−5.3%). The stability of this ordering across seeds provides confidence that the scenario contrasts reflect the pricing model's climate-conditioning rather than sampling noise.

Benchmark Comparison

At the trip level, RT-CEPIF does not outperform the static underwriting benchmark on average absolute loss error (trip MAE of 1,121 vs. 839 for B1). This is an honest limitation: individual trip-level predictions remain noisy relative to a well-calibrated static average. The empirical case for RT-CEPIF rests on aggregate policy-month performance and tail ranking, which are the levels at which premium decisions are actually made. This distinction between trip-level prediction and portfolio-level indication mirrors the argument in Fung et al. (2023) that climate exposure information is most valuable when aggregated to the contract or portfolio level rather than used as a point predictor for individual events.

At the policy-month level, RT-CEPIF materially outperforms the no-weather ablation in every scenario (Table 3). The improvement in policy-month MAE ranges from 13,424 in the baseline scenario to 29,360 in the flood-tail scenario. This means that weather information meaningfully reduces error when the task is setting adequate premium indications for books of business.

Tail Capture

The clearest evidence for RT-CEPIF's value lies in stress-tail ranking. Under the flood-tail scenario, the full climate-conditioned model captures 43.6% of the realized top-decile loss in premium-ranked policies, compared with 14.5% under the no-weather ablation, a threefold improvement (Table 4). The finding that flood-tail risk is the scenario with both the largest premium displacement and the strongest tail-ranking improvement is consistent with marine cargo insurance evidence showing that flood and storm events produce concentrated, high-magnitude claims (du Plessis et al., 2024b).

Falsification Check

A shuffled-weather falsification test assigns randomized weather labels to trips while holding all other variables constant. The full model achieves a trip-level MAE of 1,529; the shuffled model achieves 1,517. The difference is small, confirming that RT-CEPIF should not be defended as a causal weather-effect estimator at the trip level. The paper's identification claim is accordingly limited to simulation-conditioned premium indication differentials rather than structural causal recovery.

Tables

Table 1: Framework scoring coefficients used by the RT-CEPIF engine

Variable	**Frequency side**	**Severity side**
Heavy rain	$+0.34$	$+0.12$
Cold wave	$+0.16$	$+0.08$
Heat wave	$+0.12$	$+0.28$
Snowstorm	$+0.42$	$+0.22$
Flood indicator	$+0.58$	$+0.58$
Weather intensity	$+0.11$	—
Event exposure hours	$+0.017$	—
Hard-brake rate	$+0.52$	—
HOS pressure	$+0.34$	$+0.08$
Maintenance risk	$+0.25$	$+0.16$
ADAS (mitigant)	-0.20	—
Liability-only coverage	—	-0.10

Note: Coefficients are calibrated scoring weights within the RT-CEPIF synthetic engine, not regression estimates with standard errors from observed microdata. The peril ordering is consistent with crash-causation evidence from FMCSA (2007), Naik et al. (2016), and Uddin and Huynh (2020). Flood catastrophe branch: activation probability $p = 0.08 + 0.08 \times$ flood_tail_shift; severity multiplier drawn from Uniform[2.2, 4.8].

Table 2: Scenario premium indication shifts (mean indicated premium, USD)

Scenario	Small panel (3 seeds)	Seed range	vs. baseline
Baseline	$815.71	[764, 869]	—
Wet extreme	$927.21	[878, 972]	+13.7%
Heat stress	$902.50	[865, 970]	+10.6%
Flood tail	$1,248.14	[1,198, 1,308]	+53.0%
Adaptation	$774.41	[730, 810]	−5.1%
Large panel (seeds 7, 11, 19; baseline_research volume)			
Baseline	$827.99	[798, 887]	—
Wet extreme	$942.75	[910, 992]	+13.9%
Heat stress	$919.45	[891, 977]	+11.0%
Flood tail	$1,213.84	[1,122, 1,294]	+46.6%
Adaptation	$783.87	[754, 839]	−5.3%

Note: Ranges are simulation seed ranges across independently drawn panels, not sampling confidence intervals.

Table 3: Policy-month MAE: RT-CEPIF vs. no-weather ablation

Scenario	RT-CEPIF	No-weather	Improvement
Baseline	50,705	64,129	13,424
Wet extreme	53,017	76,564	23,547
Heat stress	49,163	71,349	22,187
Flood tail	66,028	95,388	29,360
Adaptation	48,634	63,143	14,509

Note: Small-panel configuration (3 seeds, approximately 25,000 trips per seed). MAE is evaluated at the policy-month aggregation level.

Table 4: Premium-ranked top-decile tail capture (flood-tail scenario)

Model	Top-10% overlap	Top-decile loss captured	Top-1% overlap
RT-CEPIF (full)	0.136	0.436	0.040
No-weather ablation	0.117	0.145	0.027

Note: Top-decile tail loss captured is the share of realized losses in the top decile of actual outcomes falling within the top decile of model-ranked policies. RT-CEPIF captures three times more flood-tail loss than the no-weather ablation.

Discussion

Implications for Insurers

The results support three specific actions for insurers with commercial trucking books. First, flood exposure should be priced as a distinct peril rather than absorbed into a general weather load. The flood-tail scenario produces premium differentials of 45 to 53% relative to baseline, and even a partial corridor-level flood indicator substantially improves tail-risk ranking. Second, policy-month premium

adequacy, not trip-level prediction accuracy, is the appropriate evaluative target for climate-sensitive rating systems. The benchmark evidence shows that weather information matters most when losses are aggregated to the level at which premium decisions are actually made, a finding consistent with the argument in Yang et al. (2023) that platform-level and route-level risk factors add the most value as portfolio management tools. Third, reinsurance and capital loads should reflect tail concentration under flood and storm regimes. Marine cargo evidence finds that higher-income markets are better positioned to price this concentration while many markets lag (du Plessis et al., 2024a); trucking insurers face the same structural challenge.

Implications for Carriers

The adaptation scenario provides quantitative evidence that operational responses to climate risk have pricing value. Carriers that invest in fleet winterization, routing flexibility, schedule slack, telematics, and cargo-handling resilience can expect lower indicated premiums. The adaptation premium ($774 to $784) is 5 to 6% below the baseline, translating to material cost savings over large fleets. Insurers should design underwriting protocols that recognize and credit documented adaptation practices rather than treating behavior as unobservable, consistent with telematics-based underwriting evidence showing that operational behavior contains information about loss risk beyond static policy descriptors (Baecke and Bocca, 2017).

Implications for Regulators

The policy implications extend beyond firm-level pricing. If static garaging territory and lagged experience systematically underprice tail risk in climate-exposed corridors, the mismatch between actual and priced risk will eventually resolve through underwriting exits, premium surges, or insolvencies rather than through gradual rate correction. Regulators should evaluate whether rate-review processes permit the use of route-time climate exposure evidence, whether disclosure requirements are adequate for explaining route-conditioned segmentation, and whether public data infrastructure for weather-freight matching needs improvement. The LTCCS data portal (FMCSA, 2024) and NOAA gridded weather products (NOAA, 2025) represent the kind of public infrastructure that can support evidence-based rate governance in commercial trucking.

Limitations

Several limitations apply. The paper's empirical design relies on synthetic data and cannot substitute for confidential real-world validation. The shuffled-weather falsification confirms that the framework should not be defended as causal weather-effect recovery. Adaptation, selection, and underwriting composition are only partially captured by the synthetic variables. Claims development, reinsurance attachment dynamics, and litigation risk are simplified relative to live insurer workflows. The framework should therefore be read as a calibrated design blueprint for what climate-sensitive trucking insurance could look like, not as a final empirical finding.

Conclusion

This paper proposes and evaluates the RT-CEPIF framework for climate-sensitive insurance pricing in U.S. long-haul trucking. The core argument is that route-time climate exposure, differentiated by peril class, carries decision-relevant actuarial information that static rating structures systematically miss. Flood-tail conditions raise indicated premiums by more than 50%, and weather-conditioned scoring triples top-decile tail capture relative to a no-weather alternative. These magnitudes are large enough to

affect pricing adequacy, capital management, reinsurance structuring, and carrier adaptation incentives.

The research builds on a growing literature linking climate hazards to insurance outcomes across transportation modes (Fung et al., 2023; du Plessis et al., 2024a,b), extends it to the specific operational and actuarial structure of long-haul trucking (FMCSA, 2007; Naik et al., 2016; Uddin and Huynh, 2020), and translates it into a pricing framework grounded in actuarial decomposition and freight-specific risk factors (Yang et al., 2023; Baecke and Bocca, 2017). The five main contributions are: formalizing the causal chain from hazard exposure to premium indication; differentiating peril effects on frequency, severity, and tail concentration; embedding falsification and robustness discipline into the empirical design; providing auditable model equations; and translating findings into insurer, carrier, and regulatory recommendations grounded in the insurance governance literature (Kunreuther and Michel-Kerjan, 2009; Jaffee et al., 2010).

Future work should migrate the framework from synthetic to observed carrier-insurer linked data, model reinsurance attachment and exhaustion explicitly under flood clustering, extend coverage to multi-line interactions, and evaluate dynamic dispatch adaptation as an endogenous response to climate exposure pricing. The institutional problem of how to price mobile commercial climate risk accurately, fairly, and transparently is unlikely to diminish as weather volatility increases.

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This research was conducted as an independent research draft. The empirical design is reproducibility-oriented, with deterministic seeds, versioned schemas, and documented scenario parameters. No confidential carrier or insurer data was used.

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